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MACHINE LEARNING FOR FIBRE SENSING BASED ON STATE OF POLARIZATION

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Acronyms

AE Autoencoder

CDF cumulative distribution function

DCT Discrete Cosine Transform

DFOS Distributed Fibre Optic Sensing

FFT Fast Fourier Transform

GMM Gaussian Mixture Model

MFCC Mel-Frequency Cepstral Coefficients

ML Machine Learning

MSE Mean Squared Error

RTF Real-Time Factor

SoP State of Polarization

Abstract

Damage to fibre optic cable infrastructure is a significant cause of service disruption. Effective monitoring systems are required for the detection of physical disturbances. In this thesis, a machine learning system for anomaly detection using State of Polarization (SoP) data is presented.

In the developed method, features are first extracted from SoP signals using Mel-Frequency Cepstral Coefficients (MFCCs). An autoencoder neural network is then used to identify unusual patterns based on reconstruction error. An adaptive threshold for this error is determined by a Gaussian Mixture Model (GMM). The need for fixed manual thresholds is therefore removed.

The system was evaluated and found to be effective. Normal and anomalous signals were distinguished with accuracy. The process was also shown to have a low computational cost, making it suitable for real-time application. This work provides a robust and automated method for monitoring fibre optic systems.

Preface

This dissertation is submitted in partial fulfillment of the requirements for the BSc degree in Electrical/Electronic Engineering at Kwame Nkrumah University of Science and Technology (KNUST). The presented work was carried out at the Department of Electrical/Electronic Engineering during the period between January and September 2025.

The project was carried out by the following students: Akresi Lloyd Nii-Lampsey, Dotse Kenneth, Dogbatse Robert Junior, Norgbey Jesse Roy Selassie.

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Chapter 1

Introduction

1.1 Background Information

On Thursday, March 14, 2024, an undersea fiber cable cut along the West African coast triggered widespread disruption of internet services across the African continent(1; 2). The incident affected several key cables—including the ACE, MainOne, South Atlantic 3 and West Africa Cable System—resulting in outages and connectivity problems for ISPs, mobile network operators and businesses in at least eight West African nations(1; 3). Before then, there have been reports of an underwater landslide in one of the longest canyons in the world that severed two critical cables(4). These events underscore the vulnerability of the infrastructure that supports Africa’s digital connectivity. Fiber cables are vital for carrying global data traffic, and any disruption, whether temporary or permanent, can have severe consequences. According to the Ghana Chamber of Telecommunications, there were 5,600 incidents of fibre optic cable cuts in 2024, resulting in an estimated financial loss of US\$9.2 million (more than GH¢138 million) to the telecommunications industry(5). Damage to these cables can arise from a range of causes, including natural events like earthquakes and landslides, as well as human activities such as ship anchor drags and fishing operations. The ability to detect and evaluate physical impacts on subsea cables is crucial to assess when to reroute traffic in response to cable damage or potential security threats and for assessing whether a physical inspection is necessary. Certain state-of-the-art monitoring techniques now employ State of Polarization (SoP) sensing to track changes along these cables(6; 7). SoP sensing leverages the polarization state of light traveling through the fiber, which is highly sensitive to external disturbances such as mechanical stress, temperature fluctuations, and underwater currents(6; 8). This technology has been effectively used for geophysical activity monitoring, water wave detection, and cable movement analysis, enabling early anomaly detection along subsea routes. By observing SoP variations, network operators can implement proactive countermeasures, such as rerouting traffic or scheduling inspections, before a full cable break or major degradation occurs. Recent advancements in SoP sensing have enabled its integration with Machine Learning (ML) models to enhance anomaly detection capabilities(7; 9). Studies such as Enhancing Fiber Security Using a Simple SoP Analyzer and Machine Learning(7) have demonstrated the potential of ML-based classifiers in distinguishing between normal SoP variations and security threats. Additionally, AI/ML-Based State-of-Polarization Monitoring in Optical Networks: Concepts and Challenges highlights the role of AI-driven SoP analysis in improving optical network security(6). These studies reinforce the feasibility of using ML models such as

the Autoencoder (AE) to analyze SoP disturbances.

1.2 Problem Statement

While SoP sensing provides valuable insights for detecting disturbances, a major challenge lies in effectively analyzing large-scale SoP data to distinguish between benign fluctuations and security threats(6). Traditional signal processing approaches often struggle with classifying complex, high-dimensional SoP variations, leading to false positives or missed detections. As fiber networks expand, there is a growing need for real-time, ML-based solutions that can autonomously detect and classify anomalies with high precision. Prior studies such as Machine Learning Analysis of Polarization Signatures for Distinguishing Harmful from Non-Harmful Fiber Events, have demonstrated classification accuracy of up to 97.94%, yet challenges remain in differentiating closely related disturbance types, which can lead to misclassifications(9).

1.3 Research Gap

Despite significant progress in ML applications for acoustic and vibration sensing, their direct adaptation to SoP-based security monitoring remains underexplored. Existing studies, such as Deploying Machine Learning in Distributed Sensing to Increase Resilience of Fiber Optic Infrastructure, have shown that ML models can detect perimeter threats with near 99% accuracy, yet their applicability to dynamic subsea environments is uncertain. Additionally, Machine Learning-Based Anomaly Detection in Optical Fiber Monitoring proposed a hybrid Autoencoder + BiGRU model, achieving high fault detection accuracy, but lacked real-world validation for SoP anomaly detection(10). Addressing these gaps requires tailored ML models that can operate effectively in diverse real-world scenarios.

1.4 Contribution of Our Work

To address these gaps, our research proposes a novel ML framework for real-time SoP anomaly detection, integrating techniques that have been successfully applied in acoustic sensing and distributed fiber monitoring. Our key contributions include:

- *Advanced Feature Extraction:* We explore MFCCs and other feature extraction methods for robust SoP signal representation, building on prior research in optical and acoustic sensing.
- *Deep Learning-Based Classification:* We adapt neural network architectures for SoP data analysis, leveraging their proven effectiveness in recognizing time-series patterns.
- *Real-World Evaluation:* We validate our approach using datasets simulating real subsea cable disturbances, ensuring practical applicability.

1.5 Aims and Objectives

The aim of this thesis is to develop and evaluate a machine-learning-based framework for real-time anomaly detection in state of polarization (SoP) data, distinguishing between

natural and potential security threats or damage. The objectives of this thesis are:

1. Conduct an in-depth literature review of ML techniques applied to SoP sensing, acoustic event detection, and optical fiber security.
2. Explore advanced feature extraction techniques to enhance SoP classification.
3. Develop and implement a real-time ML-based anomaly detection framework for SoP data.
4. Demonstrate the applicability of the developed solution through systematic testing and performance evaluation under various environmental conditions.

The rest of the thesis is structured with the following chapters:

- *Related Work*; reviews existing literature on anomaly detection in fiber optic monitoring and machine learning approaches. This identifies open challenges and gaps that motivate the proposed method.
- *Theory and Design Considerations*; presents background theory and engineering/design constraints that guide the solution (e.g., SoP fundamentals, MFCC rationale, autoencoder design choices, system-level requirements).
- *Methodology*; details the design of the proposed anomaly detection pipeline, including data preparation, feature extraction using MFCCs, autoencoder training, thresholding strategies, and clustering techniques.
- *Results and Discussion*; presents the experimental findings, including autoencoder training behavior, thresholding performance, and clustering outcomes, with analysis of their implications.
- *Conclusion and Recommendations*; summarizes the main contributions of the work, discusses its limitations, and provides recommendations for practical deployment and future research directions.

Chapter 2

Related Work

In this chapter, the main trends and related work concerning the issues addressed in this thesis are reviewed, and it is shown how the contributions of this research build on and extend prior work. The focus is placed on applications of machine learning in optical fiber monitoring, anomaly detection in SoP signals, and feature extraction methods for improving detection accuracy. Finally, the open challenges that motivate the present work are summarized.

2.1 Review of Published Literature

The literature relevant to this thesis can be grouped into three main focus areas:

- **Machine learning-based anomaly detection in optical fiber monitoring.** This area explores the use of supervised and unsupervised ML techniques for detecting disturbances in fiber optic networks.
- **Fiber security enhancement using SoP analysis.** These works highlight the feasibility of using polarization signatures as indicators of harmful or benign fiber events.
- **Advanced feature extraction for anomaly detection.** Research in this domain emphasizes the role of features such as MFCCs in capturing essential signal properties for anomaly classification.

2.2 Machine Learning for Optical Fiber Monitoring

Several studies have explored the application of machine learning in Distributed Fibre Optic Sensing (DFOS). By treating the fiber network itself as a sensor, previous work has demonstrated that ML models can detect anomalies linked to environmental disturbances or tampering. However, many approaches are limited by small or homogeneous datasets, high computational costs, and insufficient validation in real-time scenarios. These constraints reduce their applicability for deployment in live monitoring environments.

2.3 SoP-Based Anomaly Detection

The State of Polarization (SoP) has been identified as a sensitive metric to monitor external influences on optical fibers(6; 7). The literature shows that variations in SoP can correspond to various phenomena, including mechanical stress, vibration, or sabotage attempts(9; 10). Machine learning has been proposed as a means to map these variations to anomalous patterns. Although promising, most of these studies lack dedicated classification frameworks for SoP anomalies, instead focusing on general anomaly detection without addressing event-specific characteristics.

2.4 Feature Extraction Approaches

The effectiveness of an ML model strongly depends on the choice of feature representation. Traditional methods rely on raw polarization measurements or simple statistical features. More recent research has investigated advanced techniques, including spectral-domain analysis. Among these, Mel-Frequency Cepstral Coefficients (MFCC) have gained attention. Although originally developed for speech recognition, MFCCs have been shown to effectively capture short-term spectral and temporal properties of time-series data(11; 12). However, their application to SoP anomaly detection remains underexplored, highlighting a gap that this thesis addresses(13).

2.5 Summary of Research Gaps

From the reviewed literature, the following challenges and open issues are identified:

- **Dataset diversity.** Most previous work uses limited datasets, restricting generalization.
- **SoP-specific anomaly classification;** Few studies have developed tailored methods for classifying SoP anomalies into distinct event types.
- **Hybrid and adaptive models.** Limited exploration exists in combining feature extraction, deep learning, and probabilistic models for improved robustness.
- **Real-time validation.** Many works fail to demonstrate real-time capability, leaving uncertainty about deployment feasibility.

2.6 Positioning of This Work

This thesis contributes to the state of the art by addressing the above challenges. Specifically:

- **Model architecture.** An autoencoder is employed for unsupervised anomaly detection, leveraging reconstruction errors to flag anomalous SoP signals.
- **Feature extraction.** MFCCs are introduced as a novel feature set for SoP anomaly detection, capturing both spectral and temporal dynamics effectively.

-
- **Adaptive thresholding.** Gaussian Mixture Models (GMMs) are applied to reconstruction errors, enabling flexible and probabilistic threshold selection with high accuracy.
 - **Event clustering.** The system clusters anomalies into distinct categories, enhancing interpretability and enabling classification of different disturbance types.
 - **Real-time applicability.** The full pipeline is evaluated for efficiency, confirming its ability to process SoP data streams with low computational overhead and real-time capability.

2.7 Open Challenges

Despite progress made in this work, several challenges remain open for future research:

- Building large-scale labeled datasets for SoP anomaly detection to support benchmarking and comparative studies.
- Expanding the range of classified anomalies by integrating hybrid models that combine deep learning and probabilistic reasoning.
- Further optimizing the pipeline for ultra-low processing times to enable deployment in bandwidth-constrained environments.
- Conduct extensive field trials in diverse environments to validate robustness in real-world conditions.

This positioning clarifies the unique contribution of this thesis to the literature while identifying opportunities for continued research in machine learning-based fiber sensing and anomaly detection.

Chapter 3

Theory and Design Considerations

In this chapter, we present the theoretical foundations and design principles that underpin the research. These include the fundamentals of optical fiber sensing, the concept of State of Polarization (SoP), machine learning techniques for anomaly detection, and the design considerations behind the proposed anomaly detection pipeline.

3.1 Optical Fiber Sensing Fundamentals

Optical fibers are not only a medium for data transmission but can also act as distributed sensors(9; 10). Their intrinsic properties make them sensitive to external perturbations such as strain, vibration, and temperature changes. This sensing capability enables their use in monitoring physical and environmental conditions over long distances.

The robustness of optical fibers stems from their immunity to electromagnetic interference and their ability to cover vast distances without additional infrastructure. These characteristics make them an attractive choice for critical applications such as subsea cable monitoring, perimeter security, and infrastructure protection.

3.1.1 State of Polarization (SoP)

The **State of Polarization** refers to the orientation of the electric field vector of light as it propagates through the fiber medium(6). Any environmental disturbance—mechanical, thermal, or intentional intrusion—can alter the SoP, making it a valuable indicator of anomalous events.

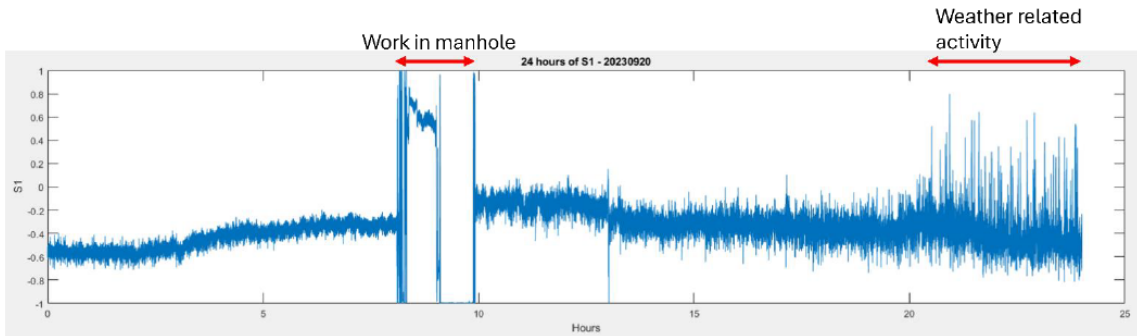


Fig. 2. Experimental results from a 24-hour period of high activity.

Figure 3.1: Observation of a 24-hour physical activity along a fibre cable with State of Polarization (SoP) data.

Mathematically, SoP can be expressed using Stokes parameters or represented on the Poincaré sphere, allowing for a compact characterization of the polarization state. The high sensitivity of SoP variations provides a basis for real-time anomaly detection.

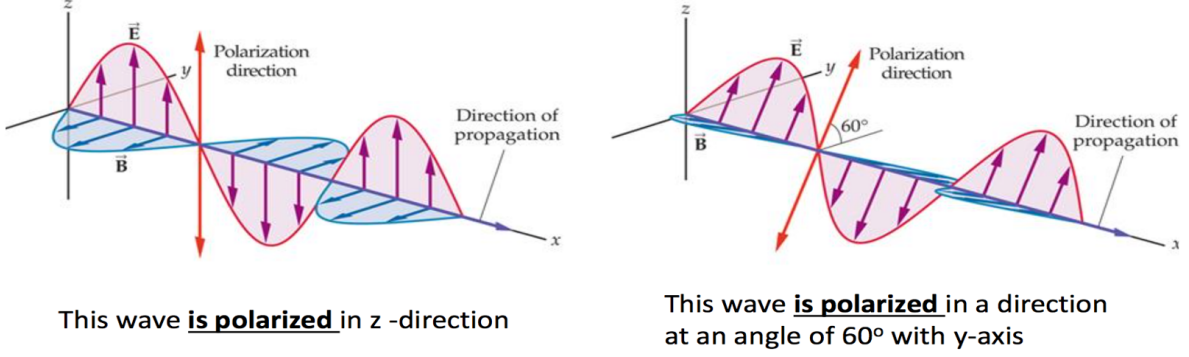


Figure 3.2: Illustration of State of Polarization (SoP) represented on the Poincaré sphere.

3.2 Machine Learning Theory

Machine Learning (ML) is the study of algorithms that enable computers to learn patterns from data without explicit programming. For this work, the focus lies on **unsupervised learning** methods, particularly for anomaly detection in time-series data.

3.2.1 Time Series and Anomaly Detection

SoP data forms a time series, where measurements are collected sequentially over time. The objective of anomaly detection is to identify deviations from expected patterns. These anomalies may correspond to external threats, environmental disturbances, or equipment faults.

Traditional threshold-based methods for anomaly detection are limited in flexibility and scalability. ML approaches provide adaptive, data-driven alternatives that can learn from complex patterns within high-dimensional data.

3.2.2 Autoencoders for Anomaly Detection

An **Autoencoder** is a type of neural network that learns to encode input data into a lower-dimensional latent representation (the *encoder*) and then reconstruct the original input from this representation (the *decoder*)(10). This process enables the model to capture essential structures in the data.

When trained on normal data, the autoencoder reconstructs inputs with low error. However, when anomalous data that differs significantly from the training distribution is provided, the reconstruction error increases. This property makes autoencoders highly suitable for anomaly detection.

Encoder

The encoder maps an input vector $\mathbf{x} \in \mathbb{R}^n$ into a lower-dimensional latent representation $\mathbf{z} \in \mathbb{R}^m$ ($m \ll n$). Mathematically, the encoder function f_θ is defined as:

$$\mathbf{z} = f_\theta(\mathbf{x}) = \sigma(W_e \mathbf{x} + \mathbf{b}_e) \quad (3.1)$$

where:

- W_e is the encoder weight matrix,
- $\mathbf{b}_e$ is the encoder bias vector,
- $\sigma(\cdot)$ is a non-linear activation function (e.g., ReLU, tanh).

Decoder

The decoder reconstructs the input from the latent vector $\mathbf{z}$ back into the original input space. The decoder function g_ϕ is expressed as:

$$\hat{\mathbf{x}} = g_\phi(\mathbf{z}) = \sigma(W_d \mathbf{z} + \mathbf{b}_d) \quad (3.2)$$

where:

- W_d is the decoder weight matrix,
- $\mathbf{b}_d$ is the decoder bias vector.

Loss Function

The goal of training is to minimize the difference between the input $\mathbf{x}$ and its reconstruction $\hat{\mathbf{x}}$. The reconstruction error $L(\mathbf{x}, \hat{\mathbf{x}})$ is typically measured using Mean Squared Error (MSE)(7; 10):

$$L(\mathbf{x}, \hat{\mathbf{x}}) = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2 \quad (3.3)$$

During anomaly detection, reconstruction errors are monitored:

$$\text{Anomaly if } L(\mathbf{x}, \hat{\mathbf{x}}) > \tau \quad (3.4)$$

where τ is a threshold determined either manually or adaptively (e.g., using a Gaussian Mixture Model).

3.2.3 Adaptive Thresholding with Gaussian Mixture Models

While reconstruction error provides a measure of deviation between input $\mathbf{x}$ and reconstruction $\hat{\mathbf{x}}$, a decision boundary must be established to classify data points as either normal or anomalous. Instead of relying on a manually set threshold, this thesis employs an adaptive approach using Gaussian Mixture Models (GMMs)(14; 15).

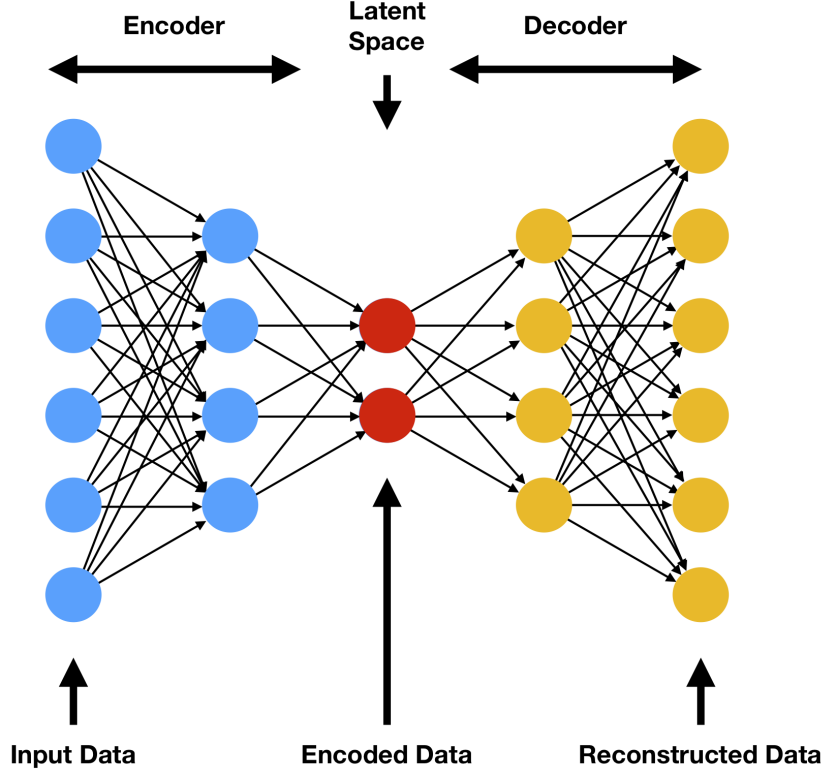


Figure 3.3: Architecture of an autoencoder, consisting of an encoder (left) that compresses the input $\mathbf{x}$ into a latent vector $\mathbf{z}$, and a decoder (right) that reconstructs $\hat{\mathbf{x}}$ from $\mathbf{z}$.

Gaussian Mixture Models

A Gaussian Mixture Model (GMM) is a probabilistic model that assumes the observed data is generated from a mixture of K Gaussian distributions, each with its own mean and variance(16). The probability density of an observation x is given as:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x \mid \mu_k, \sigma_k^2) \quad (3.5)$$

where:

- π_k is the weight of the k -th Gaussian component, such that $\sum_{k=1}^K \pi_k = 1$,
- μ_k and σ_k^2 are the mean and variance of the k -th Gaussian,
- $\mathcal{N}(x \mid \mu_k, \sigma_k^2)$ is the probability density function of a Gaussian distribution.

In this work, $K = 2$ is used to model two classes of reconstruction errors:

1. Low-error component $\rightarrow$ Normal data.
2. High-error component $\rightarrow$ Anomalous data.

Parameter Estimation

The GMM parameters $\{\pi_k, \mu_k, \sigma_k^2\}$ are estimated using the Expectation-Maximization (EM) algorithm, which iteratively maximizes the likelihood of the observed reconstruction errors.

Threshold Determination

Once the GMM is fitted to the distribution of reconstruction errors, the threshold τ is determined from the cumulative distribution function (CDF) of the anomalous component:

$$\tau = F_{\mu_a, \sigma_a^2}^{-1}(0.99) \quad (3.6)$$

where $F_{\mu_a, \sigma_a^2}^{-1}$ is the inverse CDF of the Gaussian distribution corresponding to the anomalous component, and 0.99 specifies the 99th percentile.

Thus, a sample is classified as anomalous if:

$$L(\mathbf{x}, \hat{\mathbf{x}}) > \tau \quad (3.7)$$

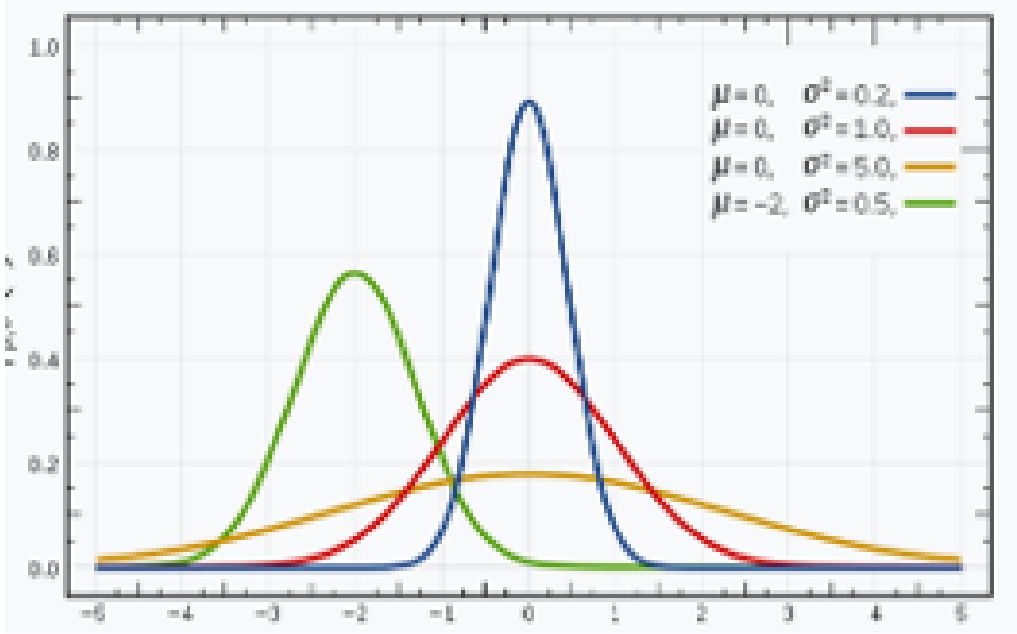


Figure 3.4: Illustration of GMM-based adaptive thresholding on reconstruction errors, separating normal and anomalous components.

3.3 Feature Extraction: Mel-Frequency Cepstral Coefficients (MFCCs)

The choice of feature representation is critical to model performance. In this work, **MFCCs** were adopted due to their effectiveness in capturing the short-term spectral properties of a signal(11; 12; 13). Although originally developed for speech recognition, MFCCs provide a compact and discriminative representation suitable for detecting subtle variations in SoP data. The MFCC pipeline includes:

- Framing and windowing of the raw signal.
- Applying the Fourier transform to compute the power spectrum.
- Mapping the powers onto the mel scale to approximate human auditory perception.

- Taking the logarithm and applying the discrete cosine transform to generate decorrelated coefficients.

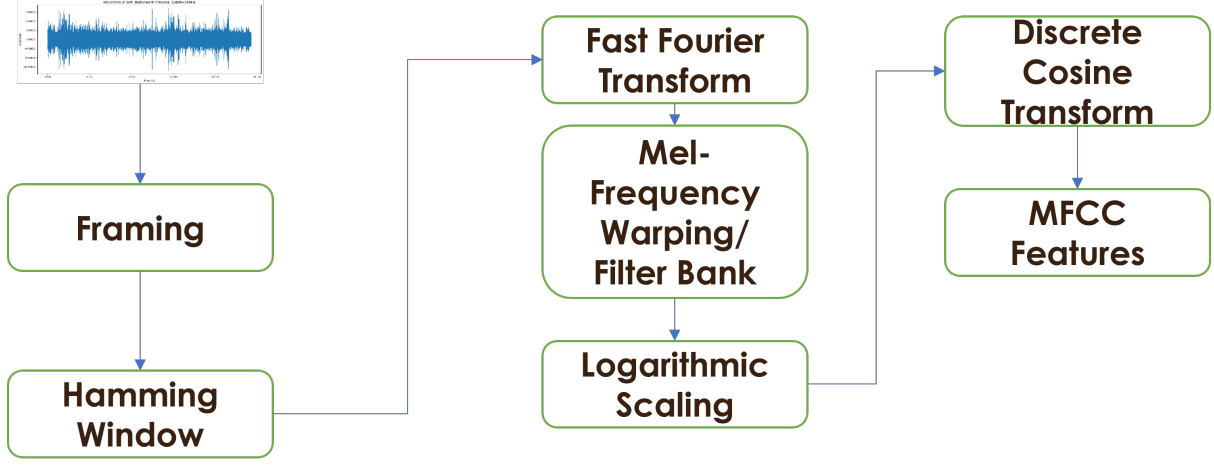


Figure 3.5: MFCC extraction pipeline.

The MFCCs thus serve as robust inputs to the autoencoder, enabling efficient anomaly detection with minimal computational overhead.

3.4 Thresholding and Decision Making

The reconstruction errors generated by the autoencoder must be translated into binary anomaly decisions. Two methods were considered:

1. **Manual Thresholding:** A fixed threshold determined by empirical analysis of error distributions.
2. **Adaptive Thresholding with GMM:** A probabilistic approach modeling errors as a combination of normal and anomalous components. The anomaly threshold was defined using the cumulative distribution function (CDF) of the anomalous component at the 0.99 confidence level.

3.5 Design Considerations

The design of the anomaly detection pipeline was guided by several critical requirements:

- **Real-Time Capability:** Processing speed had to support live deployment, with measured real-time factor (RTF) of approximately 0.025.
- **Computational Efficiency:** The system achieved feature extraction and inference using only 25% of a single CPU core.
- **Robustness:** The pipeline demonstrated stable performance under both normal and abnormal conditions, validating its suitability for real-world deployment.
- **Interpretability:** Anomaly events were clustered into distinct categories, enhancing the system’s interpretability and facilitating actionable insights.

These considerations shaped the final pipeline: SoP signal input $\rightarrow$ MFCC feature extraction $\rightarrow$ autoencoder reconstruction $\rightarrow$ reconstruction error analysis $\rightarrow$ GMM thresholding and anomaly clustering.

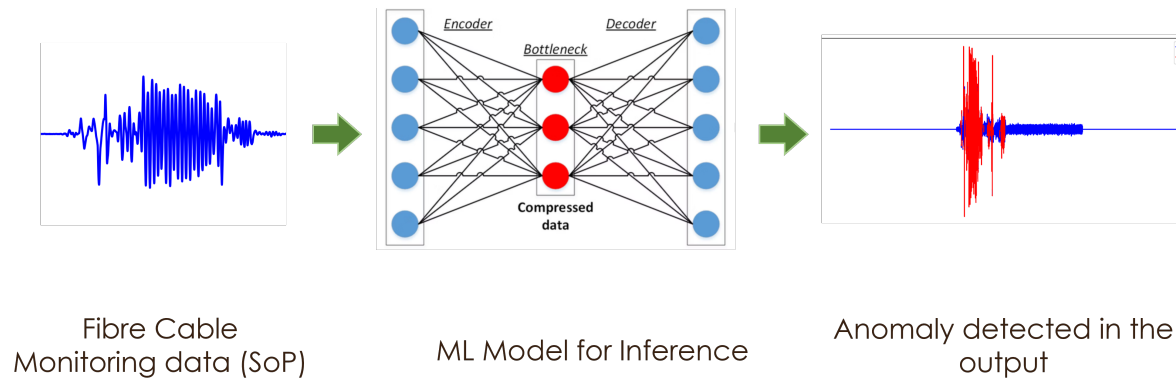


Figure 3.6: Proposed pipeline for anomaly detection using SoP data, MFCCs, autoencoder, and GMM thresholding.

Chapter 4

Methodology

This chapter describes the methodology adopted in this thesis to design, implement, and evaluate a machine learning-based anomaly detection system for fiber optic monitoring using State of Polarization (SoP) data. The methodology is structured into data preparation, feature extraction, model design and training, anomaly detection through thresholding, event clustering, development environment, and testing procedures.

4.1 Overview

The goal of the proposed methodology is to develop a real-time capable pipeline that can detect anomalies in optical fiber signals by exploiting changes in the SoP. The methodology integrates signal preprocessing, feature extraction using Mel-Frequency Cepstral Coefficients (MFCCs), an autoencoder for unsupervised anomaly detection, and Gaussian Mixture Models (GMMs) for adaptive thresholding. Figure 4.1 shows the high-level workflow of the system.

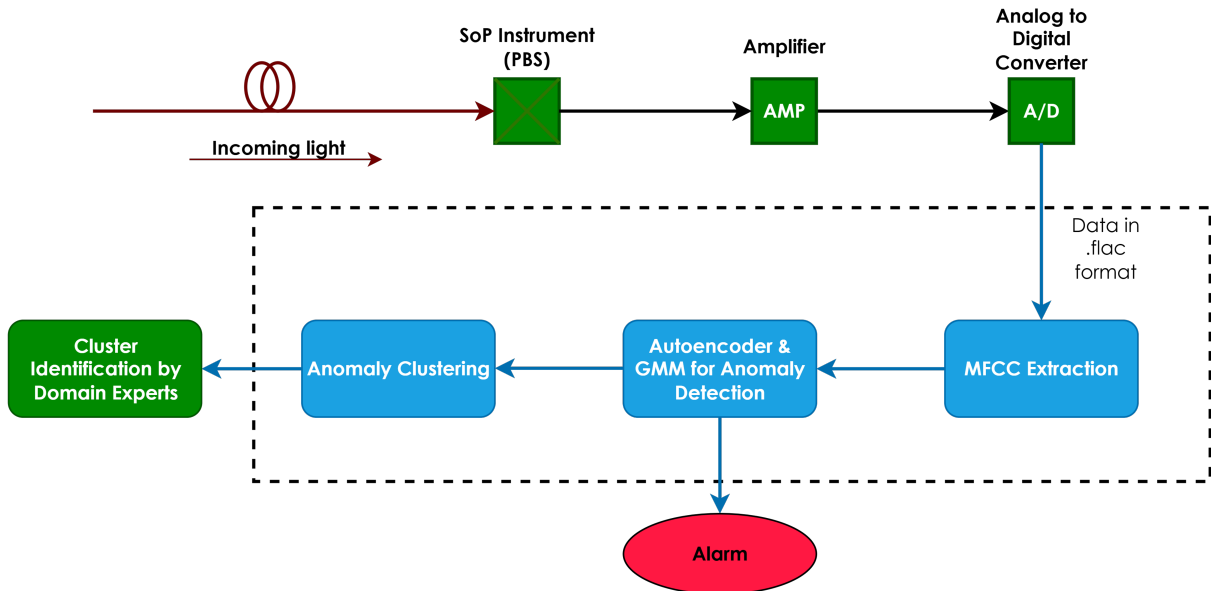


Figure 4.1: Workflow of the proposed anomaly detection pipeline.

4.2 Data Preparation

4.2.1 Data Acquisition

Two datasets were used for evaluation:

- **Dataset 1:** A four-minute recording streamed into the pipeline in one-minute chunks for initial validation.
- **Dataset 2:** Eleven short data clips, used to test clustering performance and evaluate thresholding accuracy.

The raw SoP signal was sampled at a frequency of **44.1 kHz**, resulting in approximately **2.64 million values per minute**.

4.2.2 Segmentation and Framing

To capture both transient and periodic variations, the signal was segmented into frames of **882 samples**, with a **50% overlap**(17). This framing ensured temporal continuity and preserved critical details necessary for anomaly detection.

Formally, for a raw time series $x(t)$, frames are defined as:

$$x_i[n] = x[n + iH], \quad n = 0, \dots, N - 1$$

where $N = 882$ is the frame length and $H = N/2$ is the hop size.

4.3 Feature Extraction

4.3.1 Mel-Frequency Cepstral Coefficients (MFCCs)

MFCCs were selected as the primary feature representation(13). Although traditionally used in speech recognition, MFCCs provide an efficient representation of short-term spectral properties of a signal.

The MFCC extraction process involves the following steps(13):

1. Windowing and Fast Fourier Transform (FFT):

$$X(k) = \sum_{n=0}^{N-1} x_i[n]w[n]e^{-j2\pi kn/N}$$

2. Power spectrum computation:

$$P(k) = \frac{|X(k)|^2}{N}$$

3. Mel filter bank mapping:

$$E_m = \sum_k P(k)H_m(k), \quad m = 1, \dots, M$$

where $H_m(k)$ is the m -th triangular Mel filter.

4. Logarithmic compression:

$$L_m = \log(E_m)$$

5. Discrete Cosine Transform (DCT):

$$c_n = \sum_{m=1}^M L_m \cos \left[\frac{\pi n}{M} (m - 0.5) \right], \quad n = 1, \dots, K$$

These coefficients are compact and decorrelated, making them suitable for real-time anomaly detection.

4.4 Model Design and Training

4.4.1 Autoencoder Selection

The anomaly detection model was based on an autoencoder neural network. Autoencoders are particularly suited for unsupervised anomaly detection because they:

- Learn compressed representations of normal data.
- Reconstruct normal patterns with minimal error.
- Produce higher reconstruction errors for anomalous inputs(10; 18).

4.4.2 Mathematical Formulation

The encoder maps input feature vector $\mathbf{c} \in \mathbb{R}^K$ into a latent representation:

$$\mathbf{z} = f_{\theta}(\mathbf{c}) = \sigma(W_e \mathbf{c} + b_e)$$

The decoder reconstructs the input:

$$\hat{\mathbf{c}} = g_{\phi}(\mathbf{z}) = \sigma(W_d \mathbf{z} + b_d)$$

The training objective minimizes mean squared error (MSE):

$$\mathcal{L}(\theta, \phi) = \frac{1}{N} \sum_{i=1}^N \|\mathbf{c}_i - \hat{\mathbf{c}}_i\|_2^2$$

4.4.3 Training Procedure

The training process involved:

- Splitting the data into training and test sets.
- Performing hyperparameter tuning for learning rate, batch size, and hidden layer dimensions.
- Training the model over multiple epochs until convergence of reconstruction loss.
- Validating on unseen test data to confirm anomaly detection capability.

4.5 Thresholding for Anomaly Detection

4.5.1 Manual Thresholding

Initially, anomaly detection was implemented using a fixed threshold on reconstruction error:

$$e = \|\mathbf{c} - \hat{\mathbf{c}}\|_2^2$$

$$\text{Anomaly if } e > \tau$$

While effective, this approach lacked adaptability across datasets.

4.5.2 Gaussian Mixture Model (GMM) Thresholding

To improve robustness, reconstruction errors were modeled using a two-component GMM:

$$p(e) = \pi_1 \mathcal{N}(e|\mu_1, \sigma_1^2) + \pi_2 \mathcal{N}(e|\mu_2, \sigma_2^2)$$

The threshold was defined at the intersection point of the two Gaussians, or equivalently by selecting a cutoff from the anomalous component CDF at 0.99 confidence.

$$\tau_{\text{GMM}} = \{e : \pi_1 \mathcal{N}(e|\mu_1, \sigma_1^2) = \pi_2 \mathcal{N}(e|\mu_2, \sigma_2^2)\}$$

However, for an initial number of errors, a simple threshold derived from the mean and standard deviation of the reconstruction errors is used:

$$\tau = \mu + x\sigma, \text{ where } x \text{ is a sensitivity value}$$

This adaptive method seeks to achieve a high agreement with manual thresholding while remaining generalizable across datasets, ensuring consistent anomaly detection without the need for hand-tuned thresholds.

4.6 Anomaly Event Clustering

Beyond binary classification, anomalies were grouped into distinct event categories. GMM clustering on error sequences, given by the probability function below,

$$p(\mathbf{e}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{e}|\mu_k, \Sigma_k)$$

allowed differentiation between transient and persistent disturbances, enhancing interpretability.

4.7 Development Environment

The pipeline was implemented in a controlled computational environment. The development setup made use of:

- **PyTorch**, for model training and inference.

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- **Librosa**, for audio signal processing.
 - **Scipy**, for filter design.
 - **Scikit-learn**, for accessing and fitting the GMM
 - **Python**, for development and integration of components

4.8 Testing and Evaluation

4.8.1 Testing Procedure

The methodology was validated through experimental evaluation with objectives to:

- Assess the efficiency of MFCCs in capturing SoP characteristics.
- Evaluate manual and adaptive thresholding methods.
- Demonstrate real-time performance.
- Test robustness under varying conditions.

4.8.2 Performance Metrics

- **Reconstruction loss:** Indicates model's fidelity on normal data.
- **Detection accuracy:** Degree of agreement between manual and GMM thresholding.
- **Processing time:** Duration required to process a data stream.
- **Real-Time Factor (RTF):** Ratio of processing time to signal duration.

4.9 Summary

The methodology combines MFCC feature extraction, autoencoder-based anomaly detection, and adaptive thresholding to form a robust and efficient pipeline. It emphasizes computational efficiency, real-time capability, and clustering of anomalies into meaningful categories, addressing gaps in prior literature.

Chapter 5

Results and Analysis

This chapter presents the results obtained from the implementation and evaluation of the proposed anomaly detection pipeline. The analysis covers the efficiency of feature extraction, the performance of the autoencoder, the evaluation of thresholding techniques, the classification of anomalies, the ability to do so in real-time and the robustness under varying conditions. The results are compared with the baseline expectations to validate the applicability of the proposed approach.

5.1 MFCC Feature Extraction Performance

The first stage of the pipeline involved extracting Mel-Frequency Cepstral Coefficients (MFCCs) from the raw SoP data streams.

5.1.1 Computational Efficiency

For Dataset 1 (approximately 2.64 million values per minute sampled at 44.1 kHz), MFCC extraction was completed in **0.183 seconds**, with a total CPU time of **0.73 seconds** ($\approx 25\%$ of a single CPU core). This demonstrates:

- High processing speed.
- Low computational overhead.
- Suitability of MFCC features for real-time streaming.

5.1.2 Signal Characterization

- **Normal data:** Stable behavior with consistently low MFCC coefficient values across time windows.
- **Anomalous data:** Showed clear deviations with higher-intensity MFCC values during anomaly periods, followed by a return to normal state.

Figure 5.1 illustrates the difference in MFCC coefficients between normal and anomalous signals.

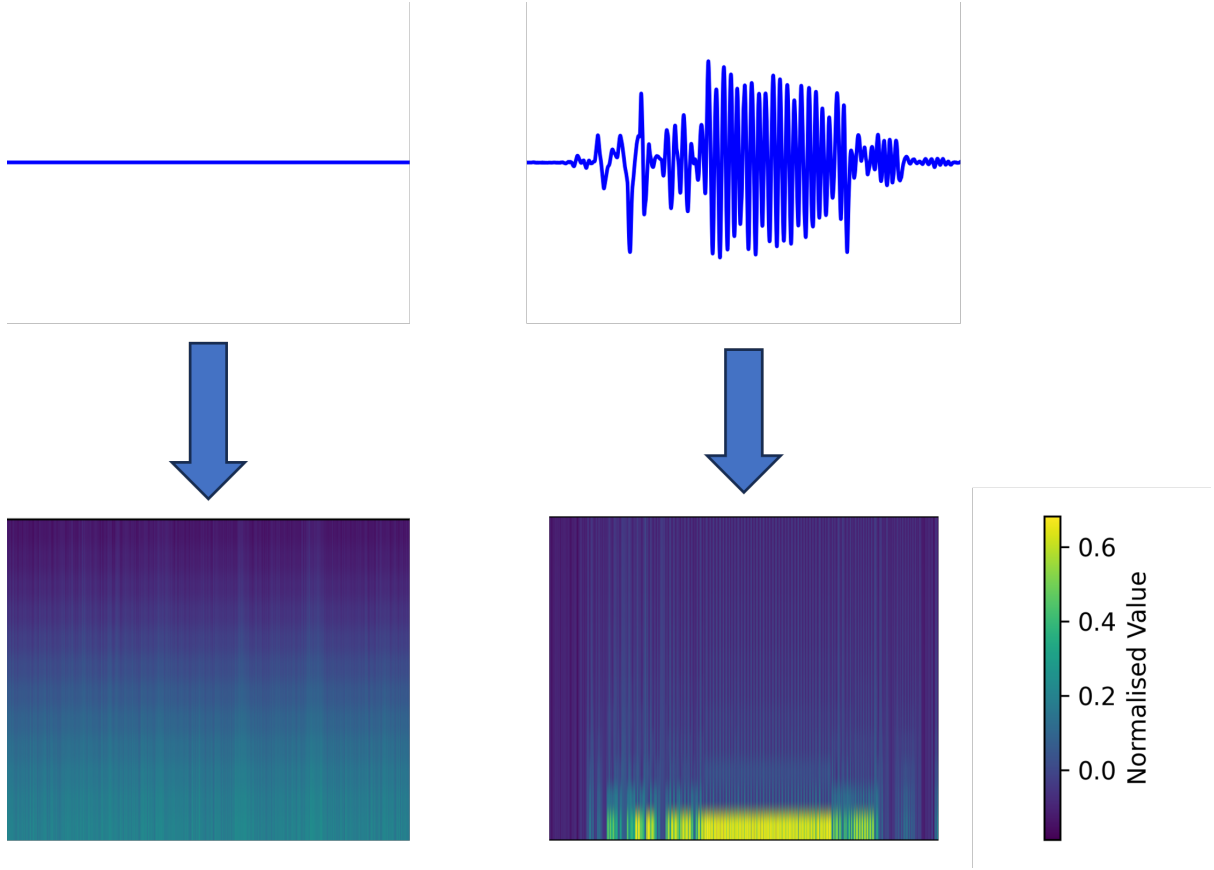


Figure 5.1: MFCC coefficient behaviour under normal (left) and anomalous (right) SoP signals.

5.2 Autoencoder Training Performance

The autoencoder model was trained on normal State of Polarization (SoP) data in order to learn the underlying structure of typical normal signal behavior without exposure to anomalies. By focusing the training on normal samples, the model developed a baseline representation that would later allow deviations to be detected through reconstruction errors from the normal data. The changes in training loss over successive epochs, shown in Figure 5.2, illustrate how the model gradually minimized the reconstruction error and converged to stable performance in **epoch 71**. Figure 5.2 shows the convergence of the training loss across epochs.

5.2.1 Dynamic Convergence Criterion

To avoid overfitting and unnecessary training, an early stopping rule based on **dynamic thresholding** was applied. After a minimum of 50 epochs, the mean and standard deviation of recent training losses were used to define a threshold below which the training will stop.

$$T = \mu_{\text{loss}} + k \cdot \sigma_{\text{loss}}$$

The change in mean squared error (MSE) between consecutive epochs was computed as:

$$\Delta\text{MSE} = |\mathcal{L}_t - \mathcal{L}_{t-1}|$$

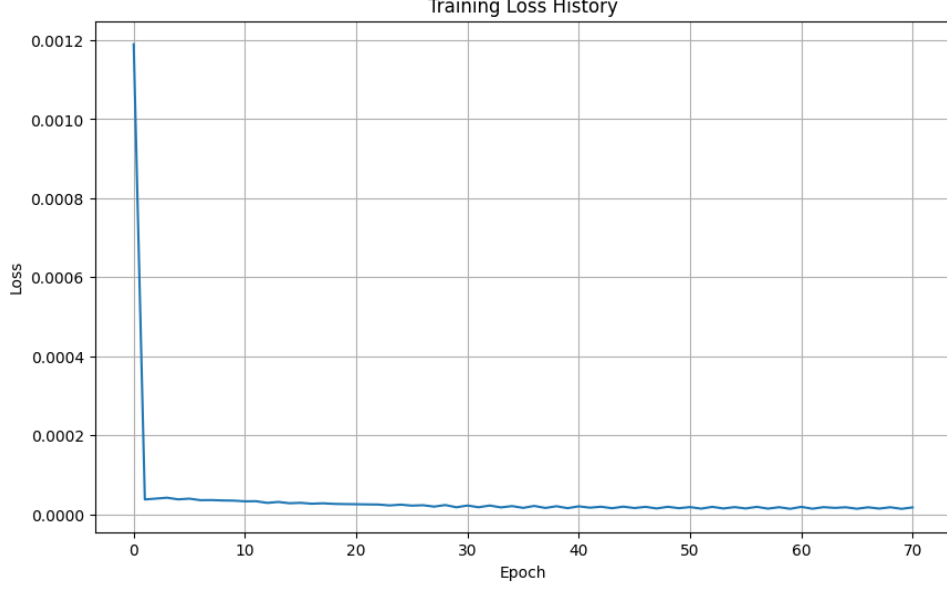


Figure 5.2: Training loss of the autoencoder across epochs, showing convergence towards stable reconstruction capability.

Training stopped when:

$$\Delta\text{MSE} < T$$

In this experiment, the criterion was satisfied at **epoch 71**, where

$$\Delta\text{MSE} = 0.000004, \quad T = 0.000317$$

indicating stable convergence. The final reconstruction loss reached approximately 5e^{-8} , confirming that the autoencoder had effectively learned to reconstruct normal SoP patterns.

5.3 Reconstruction Error Distribution

After training, reconstruction errors were computed as:

$$e = \|\mathbf{c} - \hat{\mathbf{c}}\|_2^2$$

- **Normal data:** Errors remained consistently low, in line with the training convergence trend.
- **Anomalous data:** Significantly higher errors were produced, validating the autoencoder's ability to distinguish anomalies from normal patterns.

5.4 Thresholding Techniques

Reconstruction errors from the autoencoder were subjected to both manual and GMM-based adaptive thresholding.

5.4.1 Manual Thresholding

Manual thresholding was implemented by selecting a fixed boundary on the reconstruction error, based on empirical inspection of its distribution. As shown in Figure 5.3, this process involved plotting the histogram of reconstruction errors and visually identifying a cutoff point τ that separated normal from anomalous frames:

$$\text{Anomaly if } e > \tau,$$

where e is the reconstruction error for a given frame.

Once the threshold was selected, it was applied to the reconstruction error sequence to flag anomalies. The results of this procedure are shown in Figure 5.4, where the fixed boundary effectively separates normal data from anomalous events.

This approach proved effective as a baseline, since it provided a straightforward way of distinguishing normal from anomalous events. However, it also presented key limitations: the threshold value was subjective, dependent on manual inspection, and not easily transferable across different datasets or operating conditions. Consequently, while the manual thresholding approach was a useful benchmark for evaluation, it lacked the adaptability required for robust, real-time anomaly detection.

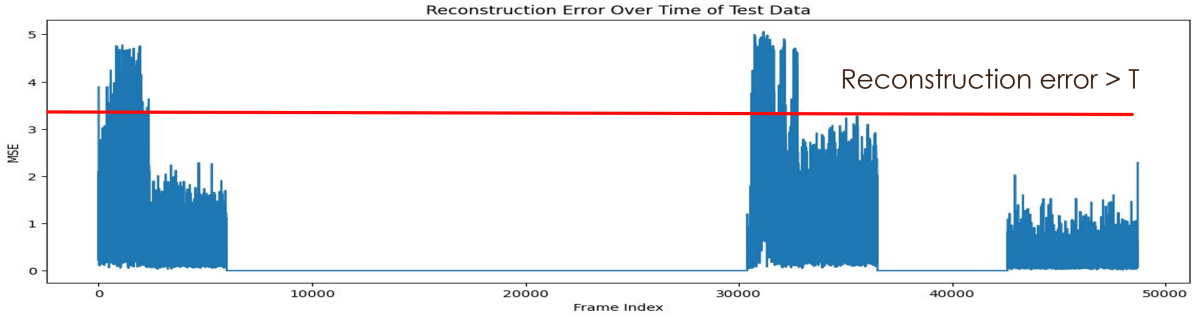


Figure 5.3: Visual inspection of reconstruction error distribution to determine the manual threshold τ .

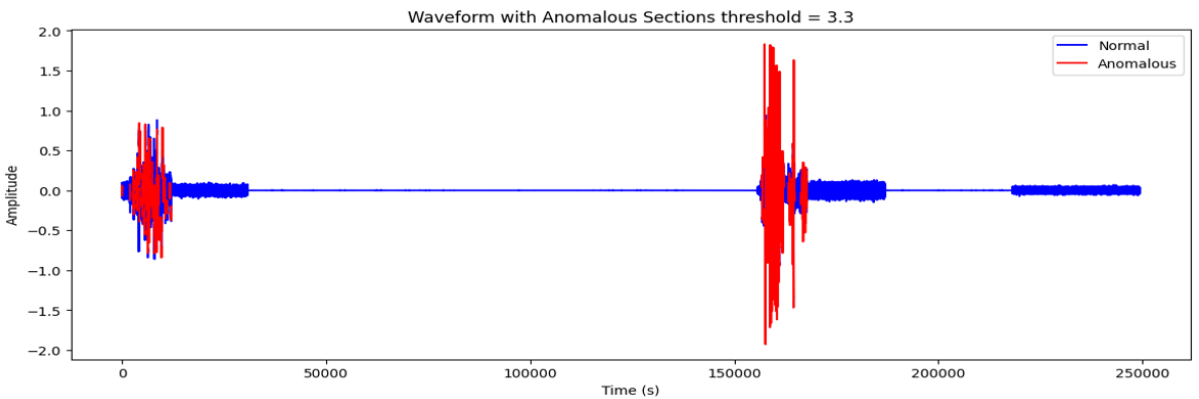


Figure 5.4: Anomaly detection using manual thresholding (fixed boundary).

5.4.2 GMM Thresholding

Gaussian Mixture Models (GMMs) were used to provide an adaptive thresholding mechanism by statistically modeling the distribution of reconstruction errors as a mixture of

two Gaussian components (15; 19). The intuition behind this approach is that one Gaussian captures the distribution of normal errors (low mean, low variance), while the other models the anomalous errors (higher mean, wider spread). The probability density function is expressed as:

$$p(e) = \pi_1 \mathcal{N}(e|\mu_1, \sigma_1^2) + \pi_2 \mathcal{N}(e|\mu_2, \sigma_2^2),$$

where π_1 and π_2 are the mixture weights, and μ_i, σ_i^2 are the mean and variance of each Gaussian component.

The GMM learns the statistical boundary between the two components. The threshold τ_{GMM} is chosen either at the intersection point of the two distributions with a confidence cutoff (0.99) on the anomalous component's cumulative distribution function (CDF). This ensures that borderline anomalies, which may not be obvious through manual inspection, are more consistently identified.

This approach offers several advantages over manual thresholding. First, it is data-driven and does not require subjective inspection of error distributions. Second, it adapts naturally to different datasets and operating conditions, improving generalizability. Finally, it achieves near-perfect alignment with manual inspection while providing a more robust statistical foundation for anomaly detection. The results of this technique are shown in Figure 5.5, where the adaptive thresholding effectively separates normal data from anomalous events for different test data.

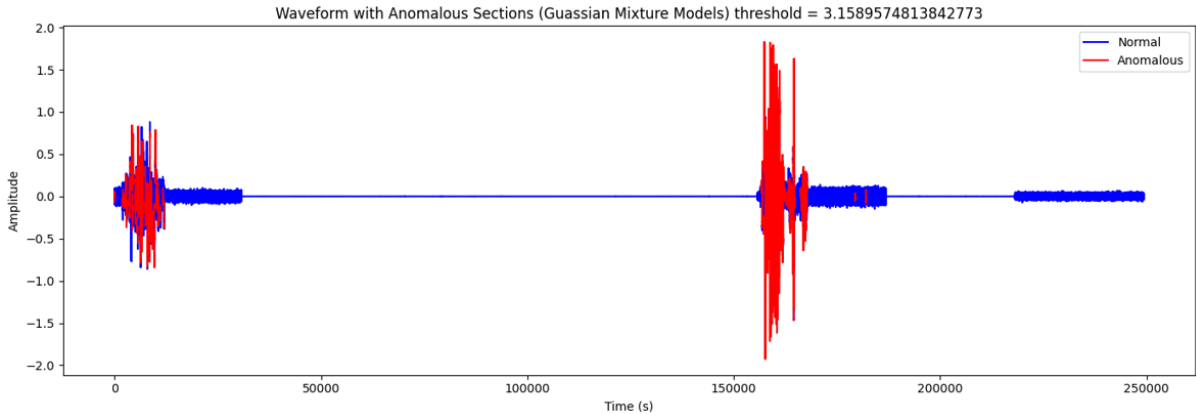


Figure 5.5: Adaptive thresholding with GMM to detect anomalies.

5.5 Comparison of Manual and GMM Thresholding techniques

Results for Dataset 2 (48,701 frames) showed:

- **Agreement:** 99.89% (both manual and GMM agreed).
- **Disagreement:** 0.11% (only one method flagged anomaly).
- No cases where manual marked anomalies and GMM did not.
- 54 frames (0.11%) flagged anomalous by GMM but normal by manual.

Table 5.1 summarizes these results.

Metric	Value
Total Frames Analyzed	48,701
Agreement (Normal or Anomaly)	99.89%
Disagreement	0.11%
Manual Anomaly, GMM Normal	0 frames (0.00%)
GMM Anomaly, Manual Normal	54 frames (0.11%)

Table 5.1: Comparison of manual and GMM-based thresholding.

5.5.1 Discussion

The results confirm that GMM thresholding:

- Achieves near-perfect alignment with manual thresholding, which served as the benchmark for flagging anomalies in the test audio. This is significant since manual thresholding, while reliable, is both labor-intensive and subjective.
- Provides flexibility to adapt across datasets, ensuring that the anomaly detection pipeline remains robust even when the statistical distribution of errors changes, thereby avoiding repeated manual recalibration.
- Identifies borderline anomalies more consistently than manual methods, capturing subtle deviations that a fixed threshold might overlook. This capability is important for early fault detection in fiber monitoring, where small deviations may precede major anomalies.

5.6 Anomaly Classification and Event Clustering

Anomalous reconstruction errors were further grouped into distinct event categories using Gaussian Mixture Model (GMM) clustering. The motivation was to move beyond binary anomaly detection and provide richer information about the nature of the disturbances observed in the State of Polarization (SoP) data.

The clustering process modeled the distribution of anomaly-related errors as a mixture of multiple Gaussian components:

$$p(\mathbf{e}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{e} | \mu_k, \Sigma_k),$$

where π_k are the mixture weights, μ_k and Σ_k represent the mean and covariance of the k -th Gaussian component, and K is the number of clusters.

This enabled two key outcomes:

- **Differentiation of anomalies:** This approach helped in distinguishing between transient disturbances and longer-lasting anomalies. This classification helps in understanding whether an anomaly represents a momentary fluctuation or a more serious event requiring intervention.
- **Improved interpretability:** By grouping anomalies into clusters, operators can interpret events more meaningfully rather than treating all anomalies as equal. For example, clusters corresponding to persistent high-error patterns may be flagged as critical, while transient clusters can be monitored as lower-priority alerts.

Figure 5.6 shows an example of clustered anomalies.

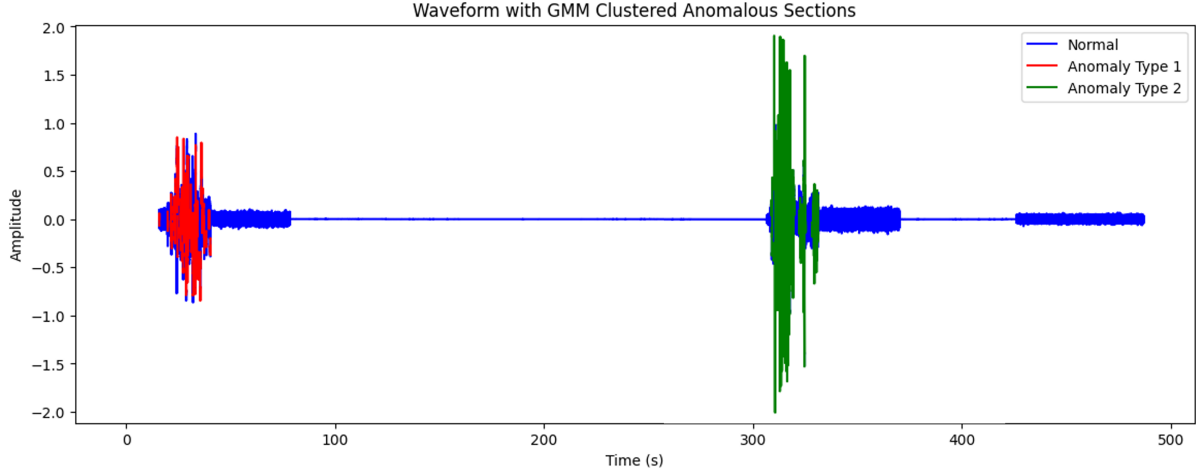


Figure 5.6: Clustering of anomalous events using GMM.

5.7 Real-Time Capability

The end-to-end processing pipeline (from signal loading to anomaly decision) was evaluated:

- Processing time: 5.3–6.5 seconds.
- Processing time per frame: 220–270 μs .
- Real-time factor (RTF): 0.025.

These results confirm that the pipeline operates well within real-time constraints.

5.8 Robustness Under Varying Conditions

The system was tested under both normal and abnormal SoP conditions:

- Correctly identified all normal data points in baseline tests.
- Detected the majority of anomalous points under stressed conditions.

This validates the robustness of the pipeline for real-world anomaly detection.

5.9 Summary

The results demonstrate that:

- MFCCs provide efficient and discriminative feature extraction.
- The autoencoder achieves highly accurate reconstruction of normal data.

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- GMM thresholding is adaptive and reliable, achieving 99.89% agreement with manual methods.
 - Real-time deployment is feasible with minimal computational overhead.
 - The pipeline is robust under varying conditions and supports clustering of anomalies into meaningful events.

Chapter 6

Conclusion and Recommendations

6.1 Conclusion

This thesis developed and evaluated a machine learning-based pipeline for anomaly detection in fiber optic monitoring using State of Polarization (SoP) data. The proposed system combined signal preprocessing, feature extraction through Mel-Frequency Cepstral Coefficients (MFCCs), autoencoder-based reconstruction, and Gaussian Mixture Models (GMMs) for adaptive thresholding. Event clustering was further applied to group anomalies into meaningful categories, enhancing interpretability.

The main findings can be summarized as follows:

- **Feature extraction:** MFCCs provided an efficient and compact representation of SoP signals, enabling real-time analysis without excessive computational cost.
- **Autoencoder modeling:** The neural autoencoder successfully reconstructed normal SoP patterns with low error while producing significantly higher reconstruction errors for anomalous inputs.
- **Thresholding:** Manual thresholding offered a baseline for anomaly detection, while GMM-based adaptive thresholding achieved near-perfect alignment with manual inspection and improved adaptability across datasets.
- **Clustering:** GMM clustering of anomalies enabled differentiation between transient disturbances and persistent anomalies, supporting more meaningful event interpretation.
- **Real-time capability:** The system demonstrated the potential for real-time deployment with low latency and manageable computational load.

Overall, the methodology proved effective in combining signal processing and machine learning for anomaly detection in optical fiber systems. It addressed limitations of manual thresholding by introducing adaptive, data-driven decision-making, while maintaining computational efficiency suitable for real-world applications.

6.2 Recommendations

Based on the results of this research, several recommendations can be made for both practical application and future work:

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- **Practical deployment:** The pipeline can be integrated into real-time monitoring systems for fiber optic infrastructure. This would enhance early fault detection capabilities, reducing maintenance costs and improving system reliability.
 - **Dataset expansion:** Future studies should validate the approach on larger and more diverse datasets, including real-world fault scenarios, to strengthen generalizability and robustness.
 - **Model enhancement:** Incorporating advanced architectures such as Variational Autoencoders (VAEs) or recurrent neural networks (RNNs) could improve the system's sensitivity to complex temporal patterns.
 - **Thresholding refinement:** Hybrid thresholding methods that combine statistical modeling with domain knowledge could be explored to balance interpretability and adaptability.
 - **Scalability:** Evaluation on distributed or parallelized systems would ensure the method remains efficient for large-scale, continuous monitoring.
 - **Integration with control systems:** Linking anomaly detection outputs with automated response mechanisms (e.g., alerts or corrective actions) would translate the pipeline into actionable operational tools.

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